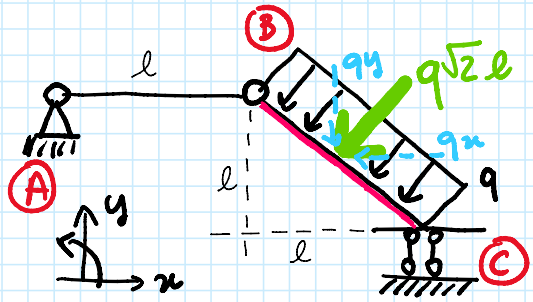
ESERCIZI di ANALISI STATICA

14 October 2022 09:13

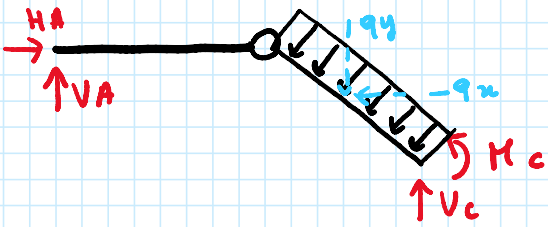
Esercizio 1



2 corpi → 6 gdl

2 cerniere + 1 doppia biella → 6 gdv

$BC = \sqrt{l^2 + l^2} = \sqrt{2}\, l$



$qx = qy = q\sqrt{2}\, l \cdot \frac{\sqrt{2}}{2} = ql$

eq. lungo asse x: $HA - qx = 0 \longrightarrow HA = ql$

" " " y: $VA + Vc - qy = 0$

eq. mom. in A: $-qx\frac{l}{2} - qy\left(l + \frac{l}{2}\right) + Mc + Vc\,2l = 0$

eq. ausiliaria (di momento in B) del corpo AB

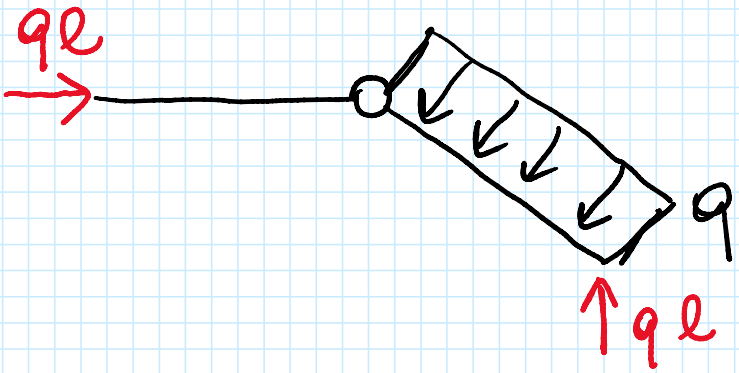
$VA\, l = 0 \longrightarrow VA = 0$

sostituisco nella terza eq. del sistema

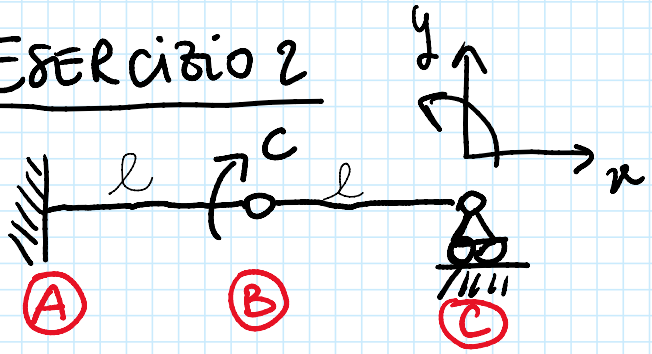
$Vc = ql$

$Mc = 0$

$VA = 0$



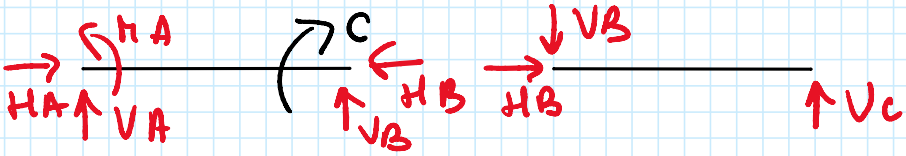
ESERCIZIO 2



2 corpi → 6 gdl

1 incastro
+
1 cerniera
+
1 carrello
→ 6 gdv

metodo 1 "struttura esplosa"



eq. corpo AB

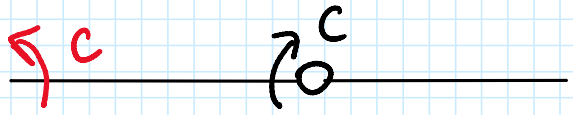
$$\begin{cases} HA - HB = 0 \\ VA + VB = 0 \\ MA - C + VB\,l = 0 \end{cases}$$

$HA = 0$
$VA = 0$
$MA = C$

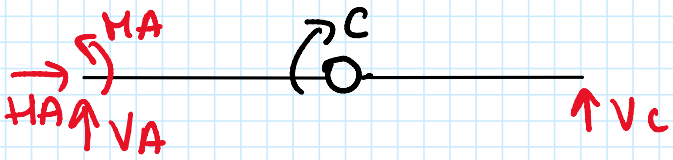
eq. corpo BC

$$\begin{cases} HB = 0 \\ Vc - VB = 0 \\ Vc\,l = 0 \end{cases}$$

$Vc = 0$
$VB = 0$
$HB = 0$



metodo 2 "eq. ausiliaria"

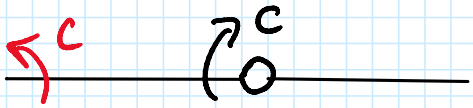


$$\begin{cases} HA = 0 \\ VA + Vc = 0 \\ MA - C + Vc\, 2l = 0 \end{cases}$$

eq. ausiliaria (eq. di momento) in B corpo BC

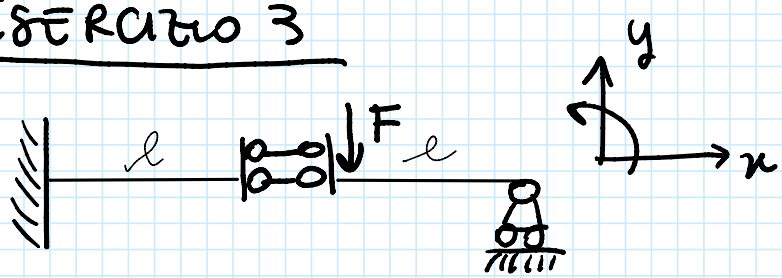
$$Vc\, l = 0 \rightarrow Vc = 0$$

$HA = 0$
$VA = 0$
$MA = C$



OTTENGO LA STESSA SOLUZIONE di PRIMA

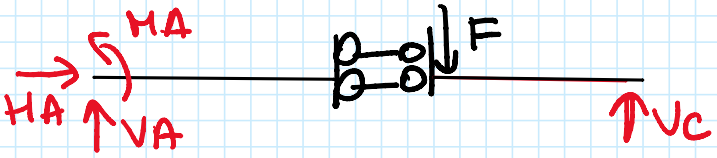
ESERCIZIO 3



2 corpi → 6 gdl

1 incastro + doppia biella + 1 carrello → 6 gdv

metodo eq. ausiliaria



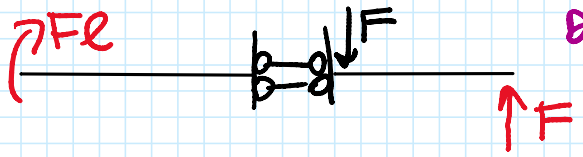
$$\begin{cases} H_A = 0 \\ V_A + V_C - F = 0 \\ M_A - F\ell + V_C\, 2\ell = 0 \end{cases}$$

eq. ausiliaria (traslazione verticale)
corpo BC

$V_C - F = 0 \longrightarrow V_C = F$

$H_A = 0$

$V_A = 0$

$M_A = F\ell - 2F\ell = -F\ell$

SEGNO "−"
DEVO CAMBIARE IL
VERSO IPOTIZZATO